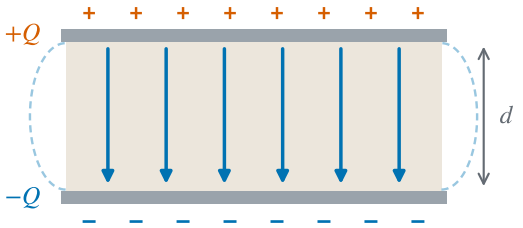
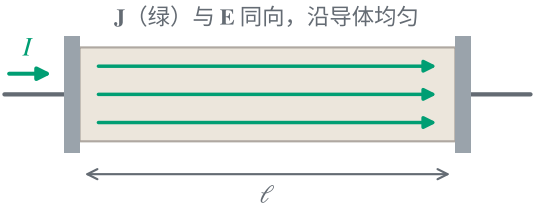
电容器：$C = Q/V$



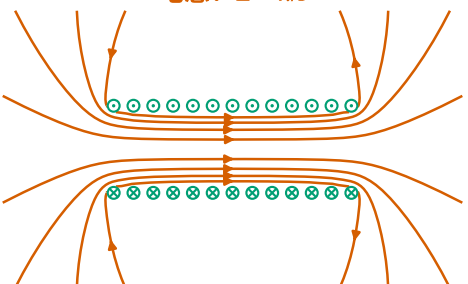
$$Q \to D = \frac{Q}{A_c} \to E = \frac{D}{\varepsilon} \to V = Ed \quad \Rightarrow \quad C = \frac{\varepsilon A_c}{d}$$

电阻：$R = V/I$



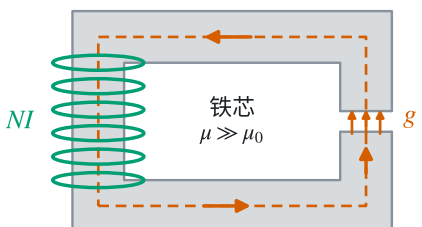
$$I \to J = \frac{I}{A_c} \to E = \frac{J}{\sigma} \to V = E\ell \quad \Rightarrow \quad R = \frac{\ell}{\sigma A_c}$$

电感：$L = \lambda/I$



$$I \to B = \frac{\mu NI}{\ell} \to \Phi = BA_c \to \lambda = N\Phi \quad \Rightarrow \quad L = \frac{\mu N^2 A_c}{\ell}$$

磁路：$\mathcal{R} = \mathcal{F}/\Phi$



$$NI = H_y\ell_y + H_g g, \ \Phi \text{ 处处相同} \quad \Rightarrow \quad \mathcal{R} = \frac{\ell_y}{\mu A_c} + \frac{g}{\mu_0 A_c}$$